

# NIHAL ERNEST

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## EDUCATION

### University of California, Santa Cruz (UCSC)

- **M.S. Computer Science** GPA: 3.85 Sep. 2024 – Mar. 2026
- **B.S. (Honors) Computer Science** GPA: 3.62 Jul. 2020 – Jun. 2024

## EXPERIENCE

### Stealth Startup

Jun. 2025 – Jun. 2026

Full Stack/AI Engineer

- Architected a full-stack platform automating real estate investment workflows, with a **Python/Flask** REST API, **SQLite** data layer, and LLM agent (**OpenClaw** + **Claude**) exposed via Slack through **MCP**, deployed on **AWS**. Directly informed **multi-million dollar real estate decisions**.
- Leveraged **SQLite** caching and parallel processing across data ingestion pipelines to reduce an 8-hour process to **under 20 minutes (~24x speedup)**.
- Designed **RESTful** API routes aggregating multi-source geospatial and third-party real estate data, with authentication, observability, and a **git-versioned prompt manager** that cut LLM iteration time and reduced inference costs through reusable, versioned prompts.

### Legal Forensics AI

Jun. 2024 – Sep. 2024

Software Engineering Intern

- Owned end-to-end development of a full-stack web app enabling legal researchers to query **millions of case documents** in natural language, with an **Angular/TypeScript** frontend and **Python** backend powered by **Llama 3** on **AWS Bedrock**.
- Replaced hours of manual keyword filtering with **LLM-driven semantic search** by integrating the Court Listener **REST API**, **surfacing relevant precedents in seconds**.
- Deployed on **AWS**, configuring cloud infrastructure for scalable document retrieval and **LLM-powered query processing**.

### Xen.AI

Jun. 2023 – Jun. 2024

Software Engineering Intern

- Built multi-tenant **Angular/TypeScript** frontends and **Python** backend function apps on **Microsoft Azure** for **enterprise pilots** in banking and healthcare, powering automated onboarding forms and admin insight dashboards.
- Integrated **OpenAI GPT-4** and **SQL** databases via **REST** endpoints, implementing **Azure** authentication and search services with **tenant-isolated access controls** to meet banking compliance requirements.

## PROJECTS

### Fishing Platform

Feb. 2026 – Present

- Built a full-stack catch-logging platform (**React** + **Python** REST API), **actively used** by anglers of all experience levels to record fish species, weight, location, and conditions into a database-backed metadata layer for historical tracking and comparison with other users.
- Developed a **recommendation engine** using historical catch data and a smart algorithm to suggest fishing plans based on time, location, gear, and target species, iterated through ongoing **user feedback**.

### Formula Forecast

Sep. 2025 – Present

- Architected a full-stack F1 prediction app (**React** + **FastAPI** + **PyTorch**) with a custom neural network (128→64→32) and a custom Huber-based loss function, achieving **89% accuracy within 3 positions**.
- Built an end-to-end data pipeline ingesting race telemetry via the **FastF1 API**, engineering 9 features per driver-race, and serving predictions through **REST** endpoints, using **time-aware cross-validation** to prevent data leakage.

### Multi-functional Discord Bot

Jul. 2021 – Present

- Launched a **Python** backend serving two frontends (Discord bot + **React Native** mobile app), used by **100+ users** across **large college Discord servers**.
- Built features including moderation, music playback, meme generation, real-time game stats / weather (via web scraping and REST APIs), and **Llama 3** chat enhanced by **web-based RAG** with Stable Diffusion image generation.

## TECHNICAL SKILLS

**Languages** Python, JavaScript, TypeScript, Java, C/C++ , HTML/CSS, SQL (PostgreSQL, MySQL, SQLite)

**Frameworks** React, Angular, React Native, FastAPI, Flask, PyTorch

**Cloud & Tools** AWS, Azure, Git, GitHub Actions (CI/CD), Linux, REST APIs, OpenAI, Llama, Supabase, Selenium